UTION



GREEN WHEELS Electric autorickshaws at a Piaggio depot in Delhi

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> **"THESE ARE THE BIGGEST POPULATION OF ELECTRIC VEHICLES RIGHT NOW. IT WAS NOT MAINSTREAM EVEN FIVE YEARS AGO. SO, THAT'S A LARGE OPPORTUNITY FOR US"**
>
> **-PULKIT KHURANA**
> *Co-founder, Battery Smart, Delhi*

e-rickshaw manufacturers registered on the Vaahan database of the Union ministry of road transport & highways—many of them assemble these vehicles after procuring components. While an exact count of these vehicles isn't available, e-rickshaws are estimated to account for over 90 per cent of the roughly 8,50,000 electric three-wheelers registered in India in the five years since 2018. The cost has been the crucial factor driving sales—an e-rickshaw can be bought for around Rs 1.20 lakh-Rs 1.30 lakh, half of what a regular autorickshaw is priced at. Bringing them into the regulatory ambit meant that auto-financing companies also opened up to e-rickshaw purchases. Now, these vehicles are a firm presence across northern cities—in the National Capital Region, for instance, e-rickshaws provide vital last-mile connectivity at the Delhi Metro stations.

"These are the biggest population of electric vehicles right now. So, that's a big opportunity for us," says Khurana, pointing out that the lead acid batteries, which need to be replaced every six months, were not designed for electric vehicles. Hence, Battery Smart, founded in 2020, is building a business model of offering e-rickshaws Li-ion batteries that can be exchanged on the go at swap centres whenever a battery pack runs out of juice. "Lead-acid has a host of problems. It can only run 60-70 km in a day and you need to replace them every 6-9 months," says Khurana, whose start-up does battery swaps for 15,000 e-rickshaws every day across 15 cities.

"When it comes to electric vehicles, three-wheelers are a low-hanging fruit," points out Sulajja Firodia Motwani, founder and CEO of Pune-based Kinetic Green. The market, she says, is at a cusp, but there are also issues holding it back. Broadly, there are two types of vehicles: one is the basic, almost *jugaad*-type, e-rickshaw running on lead-acid batteries with speeds under 25 km/hour which falls in an industry category called L3. But this is a largely unregulated market—many vehicles on the roads are still unregistered—over which several concerns, including safety, have been raised. Besides, these are not allowed in some cities—for example in Bengaluru, where local traffic authorities fear that congestion will only worsen with the low-powered vehicles.

Then, there are electric autorickshaws, which look just like regular ICE autos with an integrated windshield, except that they are powered by Li-ion batteries and are capable of higher speeds—these are classified as L5. While the L3 e-rickshaws are a big success, the L5 is still at a nascent stage—last year, 24,174 electric autorickshaws (both passenger and cargo) were sold across the country as against 7,508 in 2021, according to FADA data. The popularity of e-rickshaws shows there is a requirement for good, last-mile connectivity using electric three-wheelers, says Motwani, whose firm makes both e-rickshaws and electric autos. "It's just that we have to transition these to better quality products that will be safer and will last longer."

Under Phase II of the government's scheme for Faster Adoption and Manufacturing of Hybrid & Electric Vehicles in India (FAME-II), both e-rickshaws and electric autorickshaws are eligible for subsidies of Rs 10,000 per kilowatt hour (KWh) provided they use advanced Li-ion batteries with warranties, adhere to safety standards and meet the make-in-India parameters. Recently, representatives from the three-wheeler industry urged the government to enhance the subsidy to Rs 15,000 per KWh—as was done for two-wheelers in June 2021—so that the price gaps will narrow, says Motwani, who is also the co-chairperson of the Electric Vehicle Task Force at the Federation of Indian Chambers of Commerce and Industry (FICCI). "After the government increased the slab, two-wheeler sales have gone through the roof," she points out, adding that it will have a similar effect by reducing the cost difference between ICE vehicles and electric three-wheelers and the price gap in L3 between lead acid and Li-ion.

"The Lithium battery vehicle is what the government is now promoting," says